

Question Paper

Exam Date & Time: 12-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

HUMAN COMPUTER INTERFACE [CSE- 4069]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the causes of human error in HCI. How can designers reduce error through interface design? (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

B) Discuss the characteristics of devices used for pointing and drawing. How does Fitts' Law apply to pointing and positioning tasks? (5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Use Norman's Execution-Evaluation cycle to analyze a user setting a timer on a microwave oven. Explain the complete steps of the model. (5)
A)

Course Outcome: "CO2" / Cognitive Level: "Application"

B) You are tasked with designing an interface for an in-car infotainment system that minimizes driver distraction. An infotainment system is a central digital system in vehicles that combines information and entertainment features. Apply appropriate interaction paradigms to propose a solution and justify your choices. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Explain the key principles of screen layout that support better usability. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

B) Given an ATM interface, explain how compliance with interface standards would improve its accessibility and consistency. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Apply appropriate expert analysis technique to evaluate the interface of a hospital patient record system. Describe the techniques, the steps involved and expected outcomes. (5)
A)

Course Outcome: "CO4" / Cognitive Level: "Application"

B) Why user support is required in HCI systems? Identify any four approaches that can be used to provide user support in online ticket reservation system. Provide appropriate reasons for the methods identified. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) What is task analysis? Give one example. Compare and contrast task analysis with system analysis (5) and cognitive models.

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) A student information system needs to be developed for a college. Apply the CUSTOM approach to (5) capture the requirements of the stakeholders of this system. Explain each step in detail.

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - CSE- 4069:HUMAN COMPUTER
Apr-May 2025 INTERFACE

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - CSE- 4069:HUMAN COMPUTER
Apr-May 2025 INTERFACE

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	6	30(60%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)